9.2 Moving to k-D Spaces: Cycle Through Dimensions

You might have the impression that we hit a dead end, and even in scientific community it certainly seemed so for a long time.

The answer came in the form of a heuristic, by the hand (and brain) of Jon Louis Bentley[2]

The idea is as brilliant as simple, and stems from the considerations that led us this far in the chapter: if we restrict to 2-D spaces, instead of splitting each region in 4 sub-regions for each point, we only perform a 2-way split, but alternating splits along vertical lines to split along horizontal lines.

For each split, we partition the points in a region in 2 groups: then, at the next split, when we choose the next pivot in each of the two sub-regions, we will use a perpendicular direction to do the next partitioning.

Figure 9.2 shows a few steps of this algorithm: you can see how for the first pivot chosen we draw a vertical line passing through the point, then for the second one we draw a vertical semi-line (we are splitting a semi-plane now, not the whole plane again!), and then a vertical (semi) line again.